

TOWARDS A STRUCTURE THEORY FOR RANK TWO KLEINIAN GROUPS.

①

Defn. A Kleinian group is a discrete subgroup of $\text{Isom}(\mathbb{H}^3) \cong \text{PSL}(2, \mathbb{C}) = \text{Möb}(S^2)$.

for Γ Kleinian: - The quotient $\mathbb{H}^3/\Gamma$ is a h.p. orbifold.
- $\exists$ an open subset $\Omega \subset S^3$ s.t. $\Gamma \curvearrowright \Omega$ discontinuously, and Ω/Γ is a (poss. dis., poss. empty) Riemann surface that is the conf. boundary of $\mathbb{H}^3/\Gamma$.

If Γ is given as a set of matrices, and those matrices have varying entries, then

Suppose $\eta: \mathbb{C}^n \rightarrow \text{PSL}(2, \mathbb{C})^m$ is a family of m -sets of $\text{PSL}(2, \mathbb{C})$ -matrices parametrized algebraically by n parameters.

The set of $z \in \mathbb{C}^n: \mathbb{H}^3/\eta(z) \cong_{\text{homeo.}} \mathbb{H}^3/\eta(0)$ is incredibly wild.

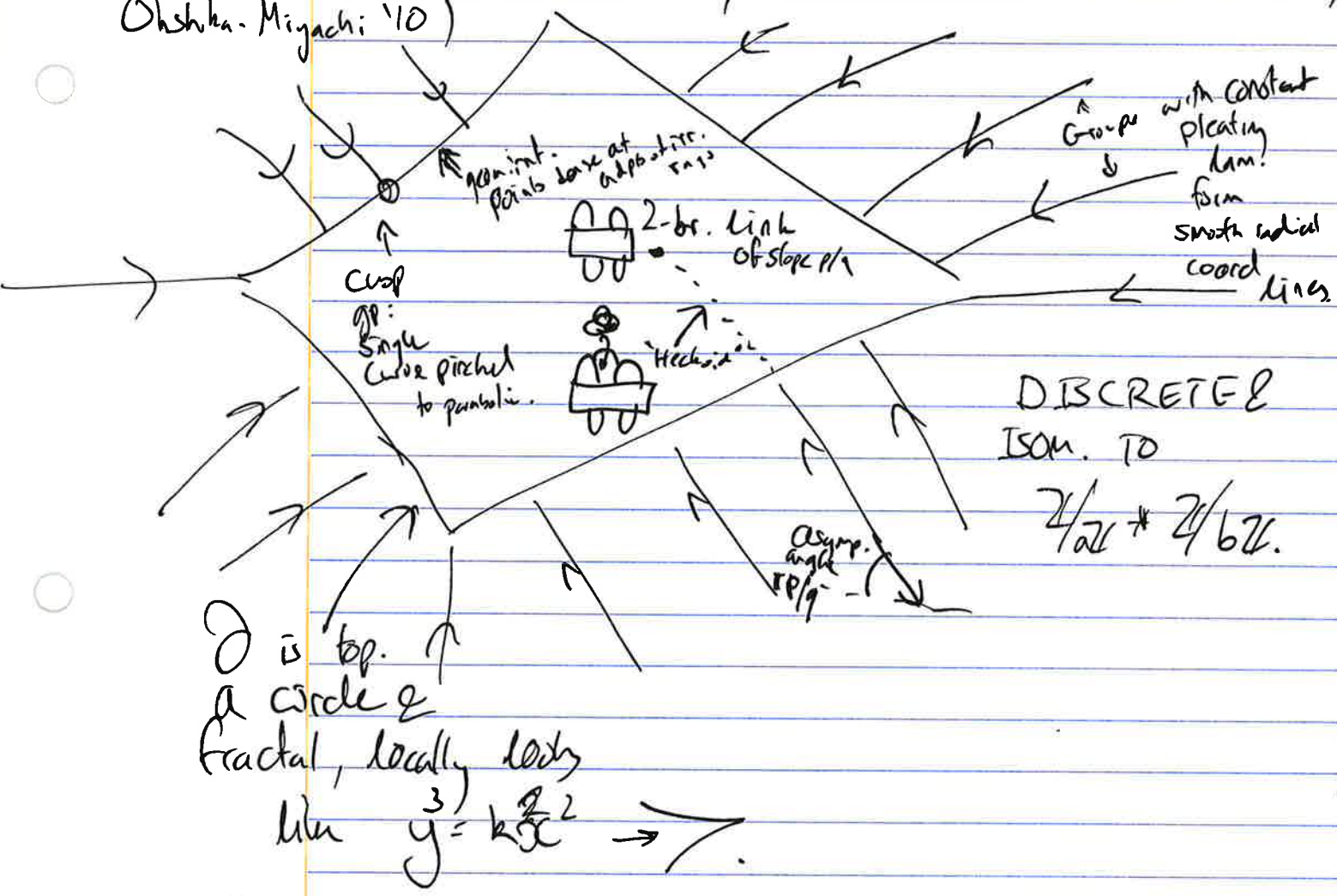
Ex. not locally con. near the boundary in $\mathbb{C}^n$ -topology.

②

We restrict to rank 2 Kleinian groups.
 Easy case: both gars either elliptic or parabolic.
 up to conjugacy, a 1-param family

$$\Gamma_z = \left\langle \begin{pmatrix} e^{i\pi/a} & 1 \\ 0 & e^{-i\pi/a} \end{pmatrix}, \begin{pmatrix} e^{i\pi/b} & 0 \\ z & e^{-i\pi/b} \end{pmatrix} \right\rangle$$

Thm (Keen-Series '14; Abiyoshi-Ohshika-Park-Sakuma-Yoshida '21;
 E.-Martin-Schillewatt '22; Chesebrough-Martin-Schillewatt (To App.),
 Ohshika-Miyachi '10)



Thm. (E.M.S.) - ^{Gons}Ja decagon s.t.

$\Rightarrow$ Complete (effectively computable!) picture of all rank 2 gars of this type.

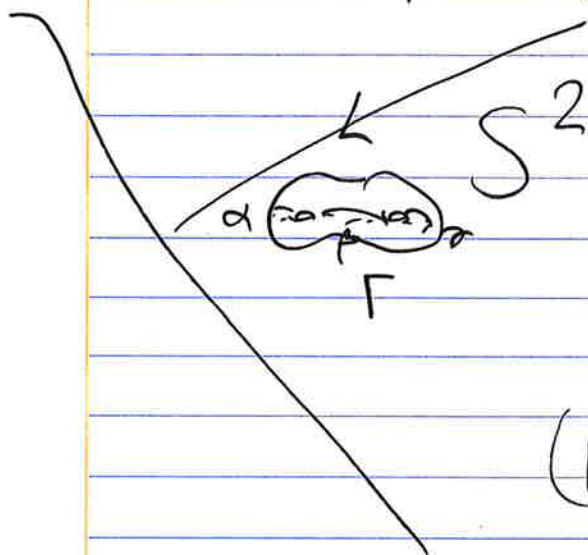
Applications: - q -rational #s have radius of conv. $\geq \frac{3-\sqrt{5}}{2}$ (E.G.M.S. AIF '26).
 - enumeration of all arith. gars in $\mathbb{H}^2$ prior to ult.

What is true for general rank 2 groups?

Naive conjectures.

- $\exists$ a Keen - Series coord. system for gens 2 schottky groups.

(Proved! E. '25 for rat. pleating rays, update soon with irr. rays).
 Note, more general than rank 2 and gives detailed info about Γ -action (fund. domain).



- If α, β, γ satisfy $tr^2 \alpha = tr^2 \beta = tr^2 \gamma$ then either

- Γ is a rank 2 max. cusp, $\Omega(\Gamma) \neq \emptyset$.

(1) - All 3 parabolic:

Γ is a rank 2 max. cusp, in particular $\Omega(\Gamma) \neq \emptyset$.

(2) - Exactly 2 parabolic:

Γ is a $\mathbb{Z}$ or $\mathbb{Z}/2$ link group in some ambient 3-sfd.

(3) - Either 0 or 1 parabolic:

Γ is a Dehn filling of a $\mathbb{Z}$ or $\mathbb{Z}/2$ link.

Thm. (1) is false in genl. (E. '26).

More precisely, you need the condition $\Omega \neq \emptyset$ as an assumption. Counterexample:



arises from a sys. of curves where all 3 are par. (1).

(ie. Fink column)

(4)

(2) & (3): Hard to prove. As a first step, in upcoming work I (E. '26-27) ~~will~~ give a topological & alg. structure theory for turn. #1 links in arb. 3-mfds extending work of (i) Cho-McCallum (ii) Sakuma - Weeks (topological only) (iii) Keen - Series, :

This relates Fun. #1

Hope $\Rightarrow$ resultg top. structures can be used to study max general solns to $tr^2 = 4$.

Can be viewed as "internal" structure theorem rank 2 gps (compare '25 work which gives structure of action on $\mathbb{C}$ - ie. 'external' structure).